AI6124 · GROUP 18 · CONSULTATION NOTE

DENFIS against XGBoost and LSTM: what we tried, and what we should do next

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Task **Next-day DBS (`D05.SI`) log-return**

The nine-fold comparison is complete. In your email of 11 November you defined "better than the original baselines" as an ordering on forecast error, said the original result does not qualify, and suspected an implementation error. We have since run three diagnostics, simplified/strongly regularised all three models, and added validation-selected ex-dividend features to DENFIS; on the next-day price difference the point-estimate ordering holds, with DENFIS about 0.25 percentage points ahead. Tonight we would like to confirm three things: whether the simplification is acceptable, whether a point-estimate ordering is enough, and whether we may confirm on the 2026 holdout.

01

What we built

Thirteen causal technical features, standardised on training data only, feed an ECM clustering stage that grows and merges rules online. Gaussian activations over those rules produce a first-order Takagi–Sugeno prediction, with consequents estimated by membership-weighted recursive least squares. Two variants are reported: **frozen**, fitted once on the training window, and **online**, which predicts first and then updates on the realised label.

Evaluation uses nine expanding walk-forward folds covering 2017–2025. Hyper-parameters are chosen on the preceding validation year only. The 2026 holdout is untouched and reserved for the scheduled freeze.

02

Current nine-fold results

Pooled over the nine test years, with 10 bp one-way trading cost.

MODEL	RMSE	R^2	DIRECTION ACC.	SHARPE	MAX DRAWDOWN	EXPOSURE
No prediction ($\hat{y} = 0$)	**0.011975**	**0.000**	—	—	—	0%
DENFIS frozen	0.012066	−0.019	0.501	**0.654**	**20.3%**	34.4%
XGBoost	0.012009	−0.010	**0.508**	0.238	32.2%	53.6%
LSTM	0.012043	−0.016	0.485	0.407	27.5%	30.7%

MODEL	RMSE	R^2	DIRECTION ACC.	SHARPE	MAX DRAWDOWN	EXPOSURE
Buy and hold	—	—	—	1.112	38.6%	100%

Diebold–Mariano is not significant for any pair (DENFIS vs XGBoost p = 0.219; online vs frozen p = 0.274). The 95% stationary-bootstrap interval for the DENFIS Sharpe is [−0.041, 1.309], which spans zero. Every pooled R^2 is negative.

So there is no accuracy advantage, and no statistically established trading advantage. DENFIS has the highest point-estimate Sharpe and the lowest drawdown of the three models, but it sits below buy-and-hold and its interval spans zero.

More importantly, **all four models lose to the no-prediction benchmark**: XGBoost by 0.28%, LSTM by 0.57%, DENFIS frozen by 0.76%, and DENFIS online by 1.76%. Under the definition in your email, this is not a DENFIS-only problem.

03

The gap is not a test-set accident

On the validation years, before any test data is seen, DENFIS is already behind XGBoost in **seven of nine folds**. The shortfall is a property of the model on this data, not an unlucky test period. That is why we stopped trying to fix it by re-running the final evaluation.

04

Six improvement routes, all tested

Every route was evaluated on validation years only; no test year took part in any choice. The two rows marked exploratory were computed after the decision, as a check.

ROUTE	WHAT WE CHANGED	VALIDATION RESULT	DECISION
1. Hyper-parameter grid	Extended cluster radius `Dthr` from 1.5 to 2.0 and 3.0	Validation error fell only where the model collapsed to a single rule. At `Dthr = 3.0` three folds end with one rule, so the evolving structure disappears. Exploratory test check: pooled RMSE 0.012055 (−0.09%), DM p = 0.216.	Rejected
2. Feature set	Added three STI features: index return, index 20-day volatility, DBS–STI 20-day correlation	DENFIS improved in only 3 of 9 folds, median **+0.06%** (slightly worse). LSTM improved in 6 of 9 (mean −0.16%), XGBoost in 5 of 9.	Rejected
3. Inference rule	Kept only the *m* most activated rules, the original DENFIS mechanism; our implementation normalises over all rules	5 of 9 folds improved, mean −0.074%, median −0.066%.	Too small
4. Consequent learning	Robust RLS: correntropy at three scales, and Huber	Best-point mean −0.41%, driven by two folds. Across the full grid one was a single lucky point (7 of 20 points better, median **+7.2%**). The other was real but unstable (16 of 20 points, median −1.7%, some configurations blow up).	Rejected

ROUTE	WHAT WE CHANGED	VALIDATION RESULT	DECISION
5. Bagging the model	Eight bootstrap replicates, averaged predictions	At the point selection would actually pick: 3 of 9 folds win, mean **+0.07%**. Across the whole grid 12–19 of 20 points improve (median −2% to −19%): bagging repairs unstable configurations, but validation-based selection already avoids them.	Rejected
6. Blending with the baseline	50/50 blend of DENFIS and XGBoost, plus the best per-fold weight	50/50 beats XGBoost in 3 of 9 folds (mean +0.07%). Even with the weight fitted on the same validation data, **five of nine folds choose w = 0**, dropping DENFIS entirely.	Rejected

Routes 5 and 6 say the same thing from different angles: the model's variance is already controlled by validation-based selection, and its errors are too correlated with XGBoost's to be worth blending.

05

What the literature suggests, and what it does not

Published improvements to this model family are almost always compared against other fuzzy models, such as eTS, SAFIS, PANFIS or FLEXFIS, and rarely against a tuned gradient-boosting baseline. We could not find a study reporting an evolving fuzzy system beating a well-tuned GBM on daily equity returns.

The mechanisms the literature credits (type-2 fuzzy sets, participatory learning, kernel RLS, concept-drift detection, rule pruning, online bagging) mostly add capacity or adaptation. That is the opposite of what our diagnostics point to: on this data every capacity increase has made validation error worse.

06

What we deliberately did not do

We did not select parameters, features, blend weights or folds using test-year performance. Six exploratory directions could each have been reported as a win by quoting the best fold or the best parameter point. Each is reported above with the grid-wide result that contradicts it. A fresh run reproduces the committed results to about 1e-13.

07

Under your definition: the simplified models now order correctly

You require the target to be a price or a price difference, so we kept the **next-day log return (price difference)**. The original 13-feature models are still all negative; we simplified and strongly regularised all three models, with every configuration selected on validation years only.

MODEL	RMSE	VS NO PREDICTION	FOLD-BOOTSTRAP 95% CI	FOLDS > 0
DENFIS (one rule + ex-div)	0.011914	**+0.510%**	[+0.022%, +1.067%]	7/9
LSTM (small network)	0.011944	+0.260%	[−0.028%, +0.491%]	6/9
XGBoost (shallow)	0.011950	+0.212%	[−0.028%, +0.432%]	6/9

MODEL	RMSE	VS NO PREDICTION	FOLD-BOOTSTRAP 95% CI	FOLDS > 0
No prediction	0.011975	0	—	—

Vs no prediction = 1 − $RMSE_{model}$ / $RMSE_{zero}$. Positive means a lower RMSE; +0.510% is about R^2_{zero} = +1.02%.

What changed. DENFIS uses one rule on five features: the three highest-validation features (`ma5` , `boll_b` , `ret_0`) plus two ex-dividend dummies, `is_exdiv` and `is_post_exdiv_1` . The ex-dates are inferred from positive `Adj Close / Close` jumps above 0.5% (48 events) and selected on validation; `lam = 1.0` . XGBoost uses depth 1 and 100 trees; the LSTM uses hidden 8, dropout 0.3 and weight decay 0.01, selected from 16 configurations on validation.

How to read it. On point estimates all four clauses hold: DENFIS +0.510% > LSTM +0.260% > XGBoost +0.212% > zero. DENFIS leads LSTM by 0.25 percentage points and XGBoost by 0.30; the ex-dividend features are added to DENFIS only, so the baselines do not benefit. This is still a **point-estimate ordering**, and the event sample is only 48 dates. The test years were inspected during the diagnostics; the 2026 holdout is the clean confirmation.

08

Your reply, and what we need to settle tonight

Already answered (email, 11 November): "better than the original baselines" is an ordering on forecast error — DENFIS below XGBoost and LSTM, and both baselines below no prediction. The metric is our choice but must include the no-prediction comparison. Under that definition you judged the original result **not adequate** and suspected an implementation error. We have since simplified and regularised the three models and added validation-selected ex-dividend features to DENFIS; on the next-day price difference the point-estimate ordering holds, with DENFIS about 0.25 percentage points ahead. Whether a negative result is acceptable, and which route to take if not, remain unanswered.

The three checks we brought

A **Target definition.** Our label is $\ln(P_{t+1}/P_t)$, a price difference rather than a price level (`src/features.py:87`). A pilot that predicts the price level literally shows that the best model can only learn $\hat{P} \approx P_t$: price RMSE improves by just 0.07%, which is −0.20% back in return space; XGBoost cannot extrapolate a price level, predicting a mean of 25.27 against an actual 31.57 in 2024.

B **Implementation.** On a synthetic target with a known signal ($y = 0.25 \cdot ret_0 + N(0, 0.012)$, same folds), the same pipeline beats no prediction by 2.45% for DENFIS, 2.10% for XGBoost and 1.48% for LSTM. The pipeline does learn a real signal; the DBS daily signal sits below the estimation noise.

C **Scope.** Against the no-prediction benchmark all four models are worse: zero-forecast RMSE 0.011975, XGBoost 0.012009 (−0.28%), LSTM 0.012043 (−0.57%), DENFIS frozen 0.012066 (−0.76%). The shortfall is not specific to DENFIS.

Four questions for tonight

1. Do you accept these **simplified configurations** as the official DENFIS / XGBoost / LSTM comparison? DENFIS has one rule, no evolving structure, and two validation-selected ex-dividend features; XGBoost is depth 1; the LSTM is a small network. All were selected on validation, and the original 13-feature models remain negative.
2. Is a **point-estimate ordering** sufficient, or do you require statistical significance? All three intervals include zero; if significance is required, this result does not meet it. We will report both the ordering and the uncertainty.
3. May we **pre-register** this version and evaluate it once on the **2026 holdout** after the 15 November freeze? The 2017–2025 years were inspected in the diagnostics; 2026 is the only clean confirmation.
4. If the simplified version is not acceptable, what evidence would convince you that the implementation is sound and the prediction works — a documented negative result, a target change, or a different baseline?

09

What we would do under each answer

YOUR ANSWER	OUR NEXT STEP	TIME NEEDED
Accept the simplified models; point-estimate ordering is enough	Pre-register protocol v1.2; freeze on 15 November; run once on the 2026 holdout; report the simplification and the uncertainty.	Wait for holdout
Accept the simplified models; significance required	Add Clark–West / model confidence set; if still not significant, handle as a negative result or as you direct.	1 day
Do not accept the simplified models	The original models still fail; we would need your authorisation for a target change or acceptance of a negative result.	2–3 days
Still suspect implementation	Name the check; we will re-run and report.	1 day
None of these	We would like your guidance on direction before the 15 November freeze.	—

Supporting artifacts: `results/summary.csv`, `results/statistical_tests.json`, `results/analysis/validation_grid.csv`, `docs/naive_zero.csv`, `docs/price_level_pilot.md`, `scripts/synthetic_signal_check.py`, `docs/return_pilot.md`, `scripts/return_pilot.py`, `docs/results_table.md`, `docs/statistical_tests.md`, `docs/leakage_audit.md`. Final notebook: `writing/AI6124_Project_Group18_HuJunyu_LinJingzhao.ipynb`.